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To the Guest Editors of the Special Issue on
Safety, Alignment, and Responsibility of Large Language Models
(Prof. Shuo Shang, Prof. Bin Yang, Prof. Christian S. Jensen, Prof. Panos Kalnis)
IEEE Transactions on Dependable and Secure Computing

Dear Guest Editors,

I am pleased to submit my manuscript, **"SENTRY: Anytime-Valid E-Process Monitoring of LLM Agent Action Streams,"** for consideration in the Special Issue on Safety, Alignment, and Responsibility of Large Language Models.

Large language model agents that call external tools are increasingly deployed with authority to move money, send messages, and act on a user's behalf, yet the prevailing runtime defenses threshold heuristic scores at a hand-picked cut-off and offer *no* statistical guarantee on how often they fire on benign behavior. This paper addresses that gap directly. SENTRY casts runtime agent monitoring as sequential change detection and builds an e-process monitored by an e-detector over per-action surprise signals, controlling the average run length to false alarm *with no independence assumption* on the highly dependent action stream—and correcting the learned reference with a distribution-free PAC threshold so the guarantee holds with finite trusted data.

The contribution is both methodological and empirical. On real trajectories collected from a capable agent (DeepSeek-V4-Flash) on the AgentDojo and τ-bench benchmarks, SENTRY detects 99% of prompt-injection attempts whose payload is observable in the black-box trace (128/198) at a 2% per-step false-alarm rate with a one-action median detection delay, using only three cheap, model-agnostic signals and no access to model internals. The study also reframes the guardrail objective from hijack-*success* prediction to injection-*attempt* detection, reports a full ablation—including three signals I implemented and rejected—and is explicit about its limitations. To my knowledge this is the first anytime-valid change-detection study on LLM-agent-safety benchmarks.

The work fits squarely within the Special Issue's scope, in particular *safety mechanisms in autonomous systems* and *adversarial robustness and attacks*: it provides a statistically controlled, model-agnostic runtime tripwire for autonomous LLM agents under indirect prompt injection, together with a reproducible evaluation on public benchmarks.

I confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. There are no conflicts of interest to declare, and no external funding was received. All code, the collected trajectory corpus, and the scripts that regenerate every reported number and figure are released with the project repository to support full reproducibility. As sole author, I am responsible for all aspects of the work.

Thank you for your time and consideration. I look forward to the reviewers' feedback.

Sincerely,

Quang-Vinh Dang